

ULAN-UDE BAIKAL, Russian Federation

WMO: 308230

Lat: 51.808N

Lon: 107.438E

Elev: 515

StdP: 95.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-33.8	-39.8	0.1	-36.3	-37.0	0.1	-33.5	9.3	-13.2	7.8	-15.5	0.5	90	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	32.1	18.0	29.9	17.6	27.9	17.0	20.0	27.6	19.1	26.3	18.2	25.1	3.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.4	22.5	16.7	12.6	21.7	15.8	11.9	21.4	59.9	27.7	56.6	26.4	53.6	25.1	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.2	-38.4	35.8	3.2	2.1	-40.7	37.4	-42.6	38.6	-44.4	39.8	-46.7	41.4		
(5)			-38.3	21.9	3.1	1.1	-40.5	22.7	-42.3	23.3	-44.1	23.9	-46.3	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-23.4	-18.4	-7.1	3.1	10.4	17.8	20.4	17.5	10.0	0.7	-10.8	-20.1	(6)
(7)		DBStd	15.84	5.92	5.76	6.47	4.35	4.31	4.07	3.14	3.34	3.93	4.50	6.22	6.37	(7)
(8)		HDD10.0	4516	1036	794	531	211	48	1	0	1	48	289	623	934	(8)
(9)		HDD18.3	6811	1294	1027	789	457	248	59	16	56	252	547	873	1193	(9)
(10)		CDD10.0	901	0	0	0	3	61	234	322	233	47	1	0	0	(10)
(11)		CDD18.3	153	0	0	0	0	2	42	79	29	1	0	0	0	(11)
(12)		CDH23.3	1943	0	0	0	3	97	603	848	363	30	0	0	0	(12)
(13)		CDH26.7	692	0	0	0	1	24	227	326	109	5	0	0	0	(13)
(14)	Wind	WSAvg	2.4	1.4	1.5	2.3	3.4	3.6	3.2	2.8	2.5	2.6	2.3	1.9	1.5	(14)
(15)	Precipitation	PrecAvg	257	4	3	2	7	14	39	70	63	29	7	9	12	(15)
(16)		PrecMax	406	10	9	8	31	43	107	139	146	54	20	24	30	(16)
(17)		PrecMin	106	0	0	0	0	1	3	14	10	6	0	2	1	(17)
(18)		PrecStd	66	3	2	2	7	11	25	33	29	14	5	5	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.1	-1.1	11.9	22.1	29.9	34.4	35.2	32.8	26.8	17.1	5.9	-2.1	(19)
(20)			MCWB	-6.4	-3.6	4.6	9.8	14.3	17.6	19.4	18.6	15.2	9.1	1.7	-3.8	(20)
(21)		2%	DB	-9.2	-4.2	7.9	18.1	26.0	31.7	32.6	29.4	23.1	13.3	2.2	-5.5	(21)
(22)			MCWB	-10.2	-6.1	2.3	7.9	12.7	16.8	18.7	18.1	14.0	6.6	-0.6	-6.7	(22)
(23)		5%	DB	-12.0	-6.8	5.0	15.0	23.1	29.2	30.2	27.2	20.5	10.4	0.2	-8.1	(23)
(24)			MCWB	-12.8	-8.4	0.5	6.4	11.5	16.4	18.5	17.4	12.7	5.2	-1.9	-9.1	(24)
(25)		10%	DB	-14.6	-9.2	2.6	12.1	20.1	27.0	28.0	25.0	18.0	8.0	-1.9	-10.7	(25)
(26)			MCWB	-15.4	-10.5	-1.0	4.9	10.0	15.9	17.8	16.7	11.3	3.9	-3.5	-11.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.5	-3.5	4.8	10.4	15.2	19.8	21.6	20.9	16.9	9.5	1.7	-3.9	(27)
(28)			MCDWB	-5.1	-1.2	11.4	21.3	27.3	28.9	30.7	28.2	23.9	16.5	5.1	-2.6	(28)
(29)		2%	WB	-10.3	-6.2	2.5	8.3	13.5	18.6	20.5	19.4	14.8	7.2	-0.3	-6.6	(29)
(30)			MCDWB	-9.4	-4.4	7.5	17.1	24.4	27.4	28.3	26.3	21.4	12.2	2.1	-5.5	(30)
(31)		5%	WB	-12.9	-8.3	0.7	6.7	12.0	17.6	19.6	18.4	13.3	5.5	-1.8	-9.1	(31)
(32)			MCDWB	-12.1	-6.8	4.6	14.4	21.6	26.3	26.9	24.7	19.6	9.9	0.1	-8.3	(32)
(33)		10%	WB	-15.3	-10.5	-0.9	5.2	10.6	16.5	18.7	17.4	11.9	4.0	-3.4	-11.4	(33)
(34)			MCDWB	-14.6	-9.3	2.3	11.7	19.1	25.2	25.7	23.3	17.1	7.4	-1.9	-10.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.5	13.8	13.8	13.5	14.5	13.8	12.4	12.3	12.8	11.7	10.2	9.6	(35)
(36)			MCDBR	11.0	14.4	15.0	19.6	20.8	18.6	17.3	17.2	18.0	16.4	11.3	10.0	(36)
(37)			MCWBR	10.1	12.6	10.0	11.2	10.1	7.3	6.5	7.5	9.6	10.3	8.8	9.2	(37)
(38)		5% WB	MCDWB	10.9	14.5	14.5	18.2	18.9	15.3	13.5	14.0	16.2	15.3	10.9	9.9	(38)
(39)			MCWBR	10.0	12.7	9.9	10.8	9.4	6.6	5.9	6.7	8.9	9.8	8.6	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.293	0.326	0.373	0.442	0.440	0.431	0.388	0.337	0.311	0.293	0.272	(40)	
(41)		taud	2.151	2.135	2.180	2.242	2.064	2.190	2.259	2.366	2.459	2.442	2.267	2.140	(41)	
(42)		Ebn at Noon	723	814	860	861	811	818	818	839	851	803	695	643	(42)	
(43)		Edh at Noon	72	98	115	123	155	138	127	109	88	73	64	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.26	2.31	3.72	4.91	5.88	6.11	5.66	4.77	3.62	2.26	1.26	0.88	(44)	
(45)		RadStd	0.07	0.11	0.18	0.27	0.33	0.35	0.33	0.23	0.33	0.16	0.09	0.07	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (11 neighbors)	N/A	N/A	-1.44	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page